Digital Platforms and Household Food Security: Understanding Access, Effectiveness, and Barriers in Mandaluyong City

Abstract

This comprehensive review explores the multifaceted relationship between the rapidly expanding digital food platform ecosystem and the complex dynamics of household food security, analyzing how these technologies influence food accessibility, nutritional outcomes, and potential systemic vulnerabilities in Mandaluyong City, Philippines. Using a descriptive-correlational design, surveys were conducted with 400 respondents through stratified random sampling. Results show that most respondents were young, low-income workers who used delivery apps only occasionally. While food insecurity was rare, mild vulnerabilities persisted. Respondents cited affordability, convenience, food quality, and digital literacy as major issues. Older and less tech-savvy groups reported greater challenges. Findings highlight that while digital food platforms provide access, barriers such as cost, usability, and trust limit their effectiveness. Policy interventions and platform improvements are recommended to enhance food security and digital inclusion.

Introduction

Digital food delivery services are rapidly shaping access to food in urban Philippines. In Mandaluyong City, dense populations and growing reliance on digital platforms make it critical to assess their role in food security. While these platforms promise convenience and access (Wu et al., 2025), concerns about affordability (Herforth et al., 2020), digital literacy (Hwang et al., 2024), and trust (Capistrano, 2021) hinder wider adoption. This study investigates how online food delivery affects household food security and identifies barriers that prevent equitable use.

Food insecurity persists in cities like Mandaluyong City. where structural barriers such as affordability, digital literacy, and socio-economic disparities affect platform usage (Capistrano, 2021; Lim, 2014). However, the extent to which our study where digital platforms contribute to Sustainable Development Goals (SDG 2: Zero Hunger, SDG 3: Good Health and Well-Being, and SDG 12: Responsible Consumption) remains unclear (Olajide et al., 2024). To fill the research gap. This study examines the digital food delivery usage, food insecurity experience, and socio-economic differences among different households. That's why there is limited research addressing localized impacts in the Philippine context, particularly among low-income households.

Globally, digital platforms have been associated with improved efficiency and accessibility of food systems (Yaiprasert & Hidayanto, 2024; Garrido & Gallardo, 2022). Yet, these benefits are not always evenly distributed. Households in lower-income brackets often struggle to sustain platform use due to high costs, unstable incomes, or limited access to reliable internet (Chungchunlam & Moughan, 2024). In the Philippine setting, the problem is compounded by digital divides, where segments of the population remain excluded from fully participating in digital economies.

Local research on digital platforms has mostly focused on consumer convenience, technological adoption, or cultural practices surrounding food choices (Wu et al., 2025). Far less attention has been given to the question of whether these platforms alleviate food insecurity or other factors, such as the high cost of living and inflation, are happening. By situating this study in the Philippines. a densely populated urban center, we capture the everyday realities of households that navigate between traditional food markets and emerging digital platforms.

This study therefore fills a crucial research gap by linking digital platform usage with household food security outcomes. By doing so, it contributes to the broader discourse on how digital transformation can support equitable and sustainable access to food in urban Philippines.

Methodology

A quantitative descriptive-correlational design was employed. Respondents (n=400) were chosen using stratified and convenience sampling from various barangays in Mandaluyong City. A validated questionnaire measured demographics, frequency of food delivery use, the Household Food Insecurity Access Scale (HFIAS), and perceptions of accessibility, affordability, and barriers. Data were analyzed using descriptive statistics, t-tests, ANOVA, and correlation analysis (Yaiprasert & Hidayanto, 2024).

Results and Discussion

Table I. Demographic Profile of Respondents

Variable	Category	Percent (%)
Age	18–34	62
Sex	Male	54
Income	Below ₱20,000	58

Table I presents the demographic profile of respondents (n=400). Most were young adults aged 18–34 (62%), predominantly male (54%), and low-income earners (monthly income below ₱20,000: 58%). This demographic trend mirrors other urban studies that show young adults and lower-income groups as primary users of digital platforms.

Table II. Frequency of Food Delivery App Usage

Usage	Percent
Daily	8.75
Weekly	32

Rarely	36.5
Occasionally	22.75

Table II shows how often people in the survey used food delivery apps. Turns out, just 8.75 percent said they did it every day. That is pretty low. Then about 32 percent went for weekly use. A bigger chunk, 36.5 percent, barely touched them. And 22.75 percent only did it now and then. Thing is, this means food delivery apps are not really part of daily life for most folks in Mandaluyong City. They seem more like an extra thing, not the main way to get meals. Probably ties into money worries, you know. People still like home-cooked food better. It is cheaper and feels more traditional. Other studies back this up too. Digital stuff like apps just adds on to old ways of cooking, especially in cities where cost matters a lot (Vedovato et al., 2022).

Table III. Household Food Insecurity Status (HFIAS)

Category	Percent (%)
Never	41
Rarely	47
Frequently	12

Table III gives a rundown on household food insecurity for the respondents. They based it on the HFIAS scale. Data points out that 41 percent of households never dealt with food insecurity at all. Then 47 percent said it happened rarely. Only 12 percent mentioned mild or occasional insecurity. Findings like this show most households stay relatively food secure. Vulnerabilities hang around though. They hit especially hard for folks on the margins. That big chunk of rarely insecure households points to something. Even stable families run into struggles now and then. Things like coming up short on money for food. Or not being able to grab the items they prefer. It all ties into this hidden food insecurity. Urban communities often miss it completely. Global research backs that up too. Food insecurity does not stick just to chronically poor households. It can touch working-class families. They might look secure on the surface. But shocks like inflation or job loss or climbing food prices leave them open (Masters et al., 2023).

Table IV. Perceptions of Digital Food Delivery

Dimensions	Mean	Standard Deviation	Verbal Interpretation
Convenience and Accessibility	1.89	0.528	Disagree
Affordability & Cost	1.94	0.485	Disagree

Service & Reliability	2.16	0.64	Disagree
Trust & Security	2.04	0.81	Disagree
Digital Barriers & Technical Issues	2.27	0.63	Disagree

Table IV presents the consolidated perceptions of respondents regarding online food delivery services, with overall responses falling within the “Disagree” range. This indicates that, in general, platforms are not fully meeting user expectations. Among the five dimensions assessed, Convenience and Accessibility ($\bar{x}$=1.89, SD=0.528) received the lowest rating, suggesting that most respondents do not view online delivery as more convenient than home cooking or grocery shopping. Similarly, Affordability and Cost ($\bar{x}$=1.94, SD=0.485) scored poorly, reflecting widespread concerns about high delivery fees and overall expenses compared to preparing meals at home. Service and Reliability ($\bar{x}$=2.16, SD=0.64) received a slightly higher mean score, but dissatisfaction remained evident, with respondents frequently reporting late deliveries and missing items. Meanwhile, Trust and Security ($\bar{x}$=2.04, SD=0.81) pointed to lingering doubts about data protection and risks of online fraud. Finally, Digital Barriers and Technical Issues ($\bar{x}$=2.27, SD=0.63) obtained the highest mean score among the five dimensions, yet still leaned negative, highlighting common struggles with app complexity, unstable internet connections, and payment-related challenges. Taken together, these results suggest that while digital food platforms offer households new ways to access meals, significant barriers remain in terms of cost, convenience, trust, and digital literacy. These findings are consistent with prior studies, which emphasize that affordability and usability strongly influence consumer acceptance of food delivery platforms (Wu et al., 2025; Chawla & Puri, 2021).

Conclusion

Online food supply in Mandaluyong city exists as an occasional contingency; findings have confirmed that the method has been used as a supplementary rather than the major source of food. An air of a young, male, low-income, and working-class life was carried out by the respondents: mostly people who were working full time but still enrolled under some program or less-skilled workers. Online food service worked only occasionally as an avenue to supplement their daily food intake rather than as an actual daily source of food, with the majority not particularly food insecure, although the mildness or occasional nature of the related challenges is an expression of underlying vulnerabilities. Respondents seemed to find food delivery services inconvenient as they emphasize the expensive costs and found them unreliable; moreover, such services seem little user-friendly. Instead of food quality, concern was raised only over usability. The real issue was less of the cost and more of the trust, digital literacy, or prior disappointment being a hindrance to actualizing these services. The carefully built attitude centered on payment security and spotty security appeared as a crosswise focus. Attitudinal factors carried forward with most younger, high-income, and highly educated customers in the transaction-set, not having empathy for the pain but wishing for a good deal of convenience and digital features; whereas the poor and less educated, mostly older female customers, somewhat more distributed, have-nots to actualize all that—convenience and technology were their biggest concern.

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